Ajloun National University · 314341 AI in Cybersecurity and Cloud Computing

Chapter 1 · Lecture 1 of 2

Introduction to Artificial Intelligence



What AI is, the AI family, and how machines learn: definitions and machine-learning models

Dr. Mohammed Tawfik

Learning Outcomes

By the end of this chapter you will be able to:

1. Define artificial intelligence and explain why it is a field of computer science.
2. Describe the four ways of defining AI and the idea of an intelligent agent.
3. Explain, in order: machine learning → deep learning → generative AI → LLMs → AI agents.
4. Describe how common machine-learning models work, using a simple example for each.
5. Classify AI systems by capability and by functionality.
6. Give examples of AI in different domains, especially in cybersecurity.

Chapter Roadmap



Part 1

What is AI?

Definition, the four ways to define AI, agents



Part 2

The AI family

ML and its models → deep learning → generative AI → LLMs → agents



Part 3

Types of AI

By capability and by functionality



Part 4

A short history

1950 to today



Part 5

Branches and domains

Where AI is used



Part 6

AI and cybersecurity

Defenders, attackers, risks

AI Is Already Around You



Spam filter

Moves junk and phishing e-mail out of your inbox.



Face unlock

Recognises your face to open the phone.



Maps

Predicts traffic and suggests the fastest route.



Chatbots

Answers questions written in plain language.

Think: which of these did you use today? All of them use artificial intelligence.

First Question: What Is Intelligence?

Before we define artificial intelligence, we need a simple idea of intelligence.

In everyday terms, intelligence is the ability to learn, understand and decide well.

Artificial intelligence tries to give machines some of these abilities.



Learn from experience



Reason and solve problems



Understand language



Perceive see and hear the world



Decide and act to reach a goal

Definition of Artificial Intelligence

> **“The science and engineering of making intelligent machines, especially intelligent computer programs.”**
> — John McCarthy, who gave the field its name (1955)

In plain words

AI is a field of computer science. It builds systems that do tasks that normally need human intelligence: recognising faces, understanding speech, translating, deciding.

AI is not one program. It is a whole area of study, like medicine or cybersecurity, with many methods inside it.

AI Is a Field of Computer Science

Computer science

- Programming
- Networks
- Databases
- Cybersecurity
- **Artificial intelligence**

AI also borrows ideas from other sciences:

- Mathematics
- Statistics
- Psychology
- Linguistics
- Neuroscience
- Philosophy

Four Ways to Define AI

	Like humans	Rationally (the best way)
Thinking	**Thinking humanly** Copy how the human mind works (cognitive science).	**Thinking rationally** Follow correct rules of logic to reach conclusions.
Acting	**Acting humanly** Behave so a person cannot tell it from a human (Turing test).	**Acting rationally** Choose the action that best reaches the goal. Used by most AI today.

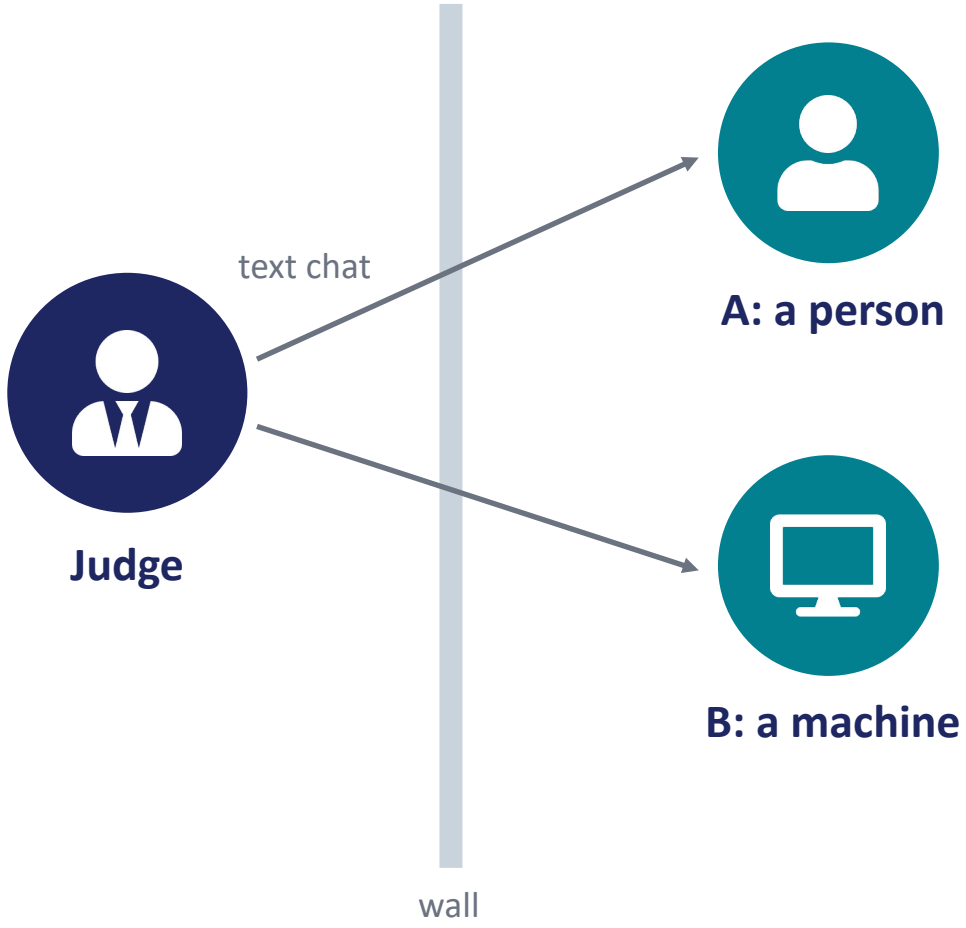
Acting Humanly: The Turing Test (1950)

Alan Turing suggested a practical test.

A judge chats by text with two hidden partners: one person and one machine.

If the judge cannot tell which one is the machine, the machine passes.

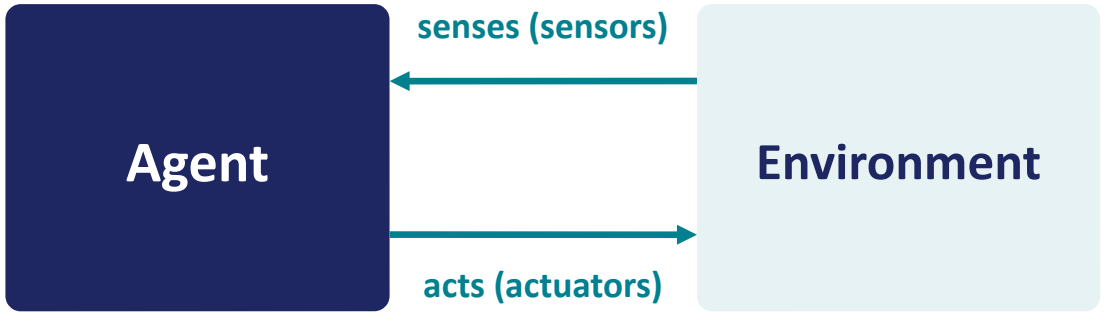
Limitation: fooling people is not the same as understanding.



Acting Rationally: Intelligent Agents

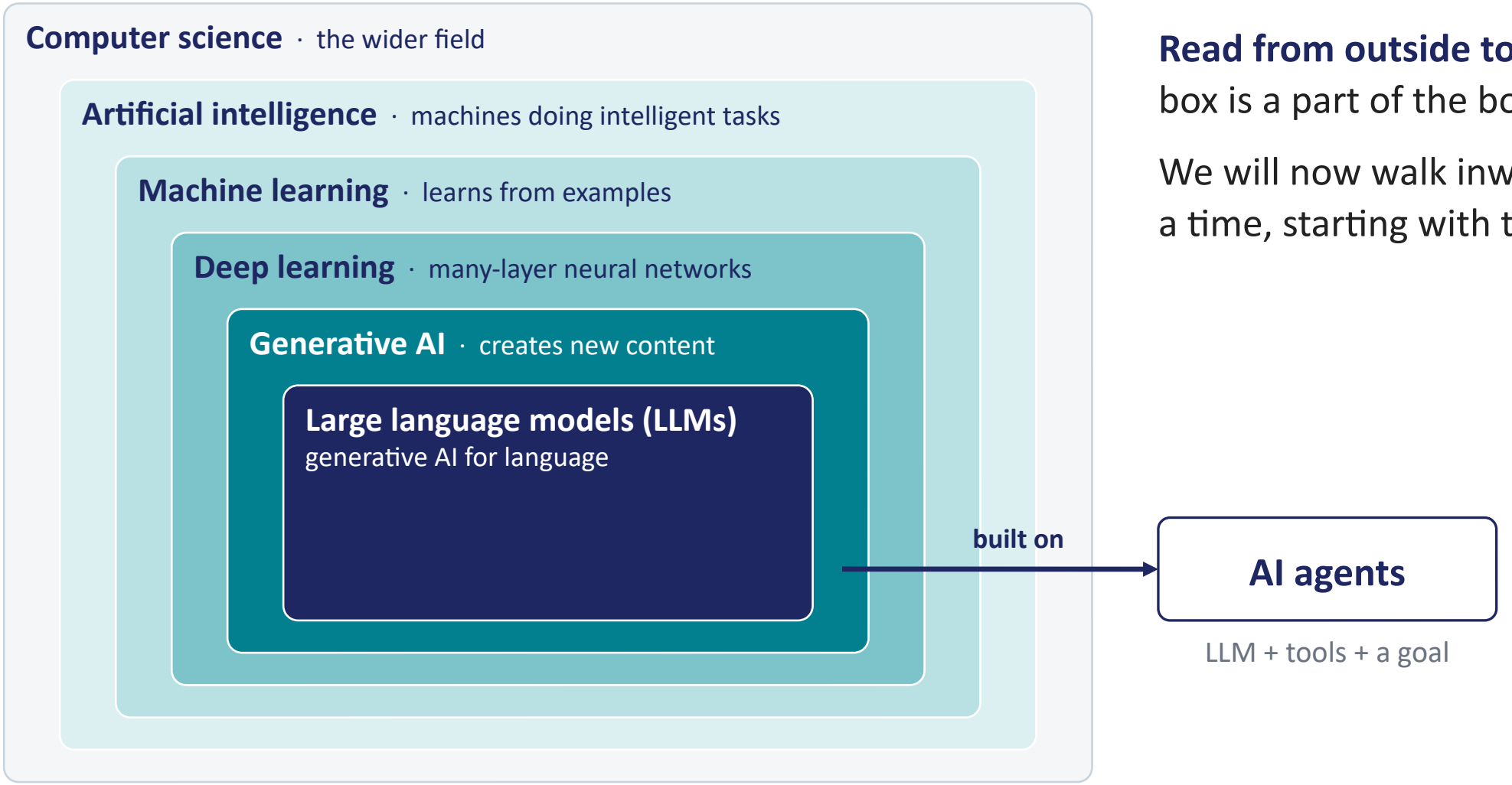
An agent is anything that senses its environment and acts on it.

A rational agent chooses the action expected to best reach its goal.



Agent	Senses	Acts
Robot vacuum	camera, bump sensor	moves, cleans
Spam filter	e-mail words, sender, links	inbox or spam folder
Self-driving car	cameras, radar	steers, brakes

The AI Family: From Big to Small



Read from outside to inside: each box is a part of the box around it.

We will now walk inward, one step at a time, starting with the oldest idea.

Before Learning: Rules Written by People

The first AI programs used rules written by human experts: IF this, THEN that.

This is called symbolic AI. Expert systems in the 1970s–80s, such as MYCIN for medicine, worked this way.

Good at: clear, logical problems such as chess.

Weak at: fuzzy problems such as recognising a face or a voice: nobody can write all the rules.

IF the e-mail says “you won a prize” THEN spam

IF the sender is in my contacts THEN not spam

IF the link hides a strange address THEN spam

Attackers change their tricks every day. Who will write the next thousand rules?

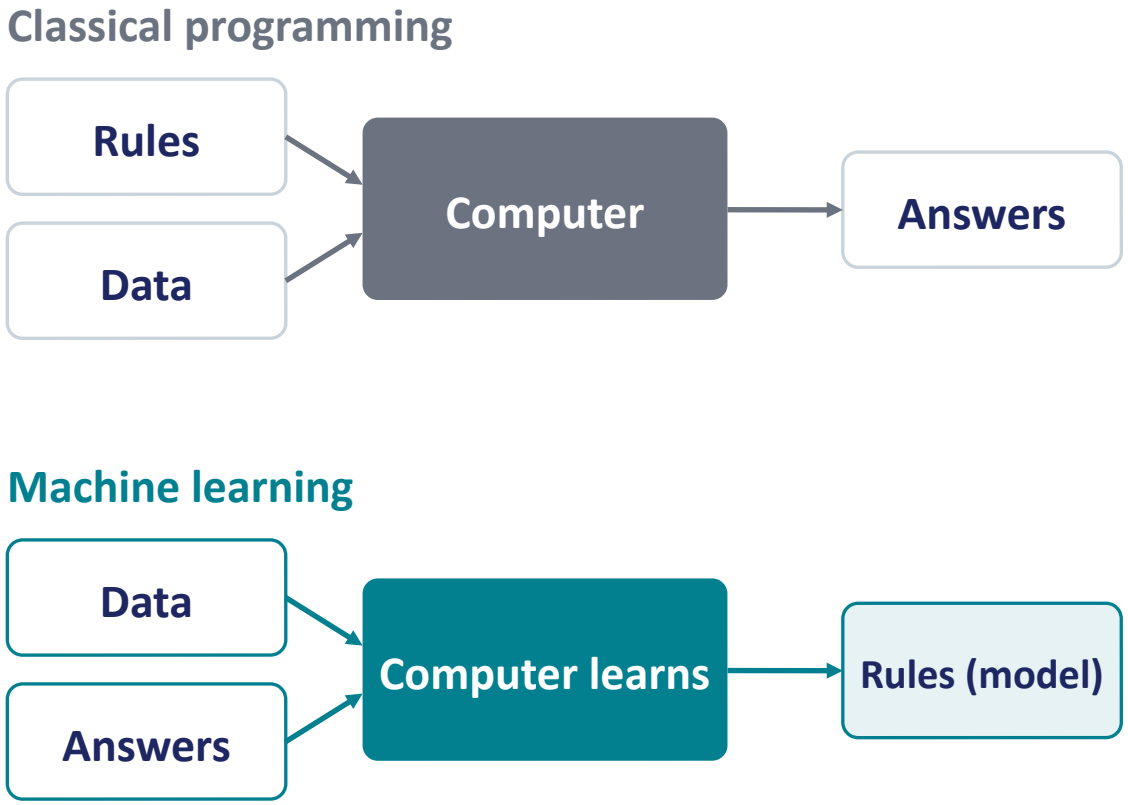
Machine Learning: Learning the Rules from Data

Instead of writing the rules, we give the computer many examples together with the right answers.

The computer finds the pattern (the rules) by itself and uses it on new data.

A machine-learning system is trained, not programmed.

Example: show it thousands of e-mails marked “spam” or “not spam”; it learns the clues.



Defining Machine Learning

Simple definition: machine learning gives computers the ability to learn from data without being explicitly programmed for every case (after Arthur Samuel, 1959).

Precise definition (Tom Mitchell, 1997)

A program learns when its performance (P) on a task (T) gets better with experience (E).

T — Task

What we want done

Spam example: Sort e-mails into spam and not spam

E — Experience

The data it learns from

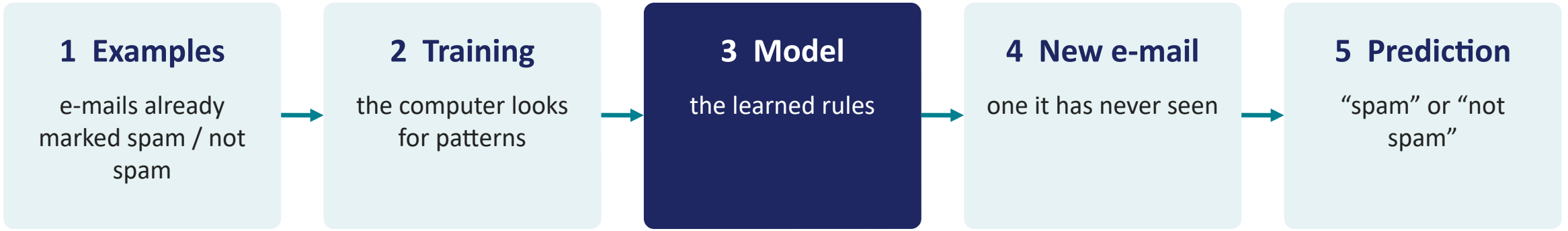
Spam example: Thousands of e-mails already marked by users

P — Performance

How we measure success

Spam example: Percentage of e-mails sorted correctly

How a Machine Learns: Train, Then Predict



Three words you will hear all semester

Data (examples)

The records the model learns from, e.g. 50 000 old e-mails.

Features (clues)

What the model looks at: words, links, sender, time sent.

Label (right answer)

The correct answer for each example: spam or not spam.

Finally we test the model on examples it has never seen, to check that it really learned.

Three Ways Machines Learn



Supervised learning

Learns from examples that have the right answer (labels).

Example: Spam / not spam; malware / clean file.



Unsupervised learning

No answers given. Finds groups and unusual items by itself.

Example: A login at 3 a.m. from a new country.



Reinforcement learning

Learns by trial and error, collecting rewards for good actions.

Example: Game players such as AlphaGo; robots.

What Do Models Predict? Classes or Numbers

Classification

The answer is a category (a class).

- Spam or not spam?
- Malware or clean file?
- Which attack type?

Regression

The answer is a number.

- House price from its size
- Exam mark from hours studied
- Network traffic in the next hour

Clustering

No answers given: find groups of similar items (unsupervised).

- Group customers by habits
- Group computers by behaviour
- Spot the odd one out

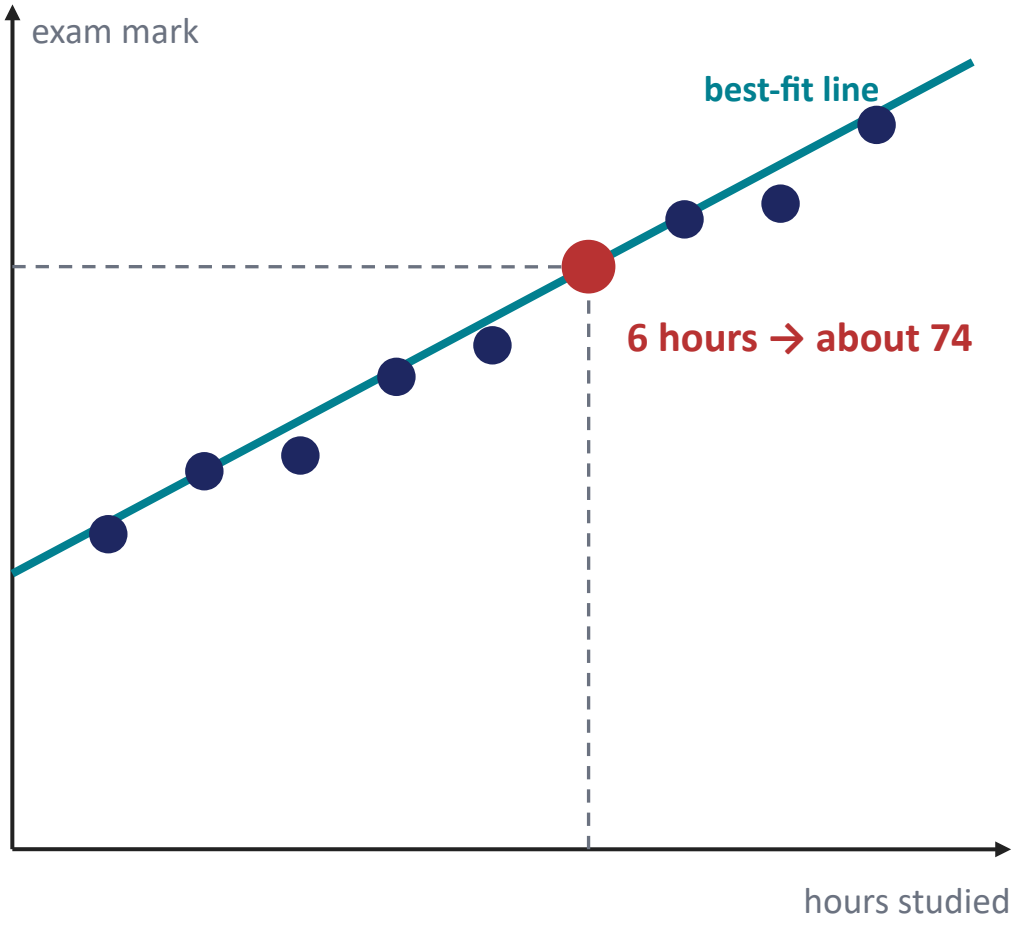
Model 1: Linear Regression

Supervised · regression (predicts a number)

What it does: draws the straight line that best fits the examples, then reads the answer for a new case from the line.

How: it chooses the line that stays as close as possible to all the points.

Simple example: hours studied → exam mark. For a student who studies 6 hours, the line predicts a mark of about 74.



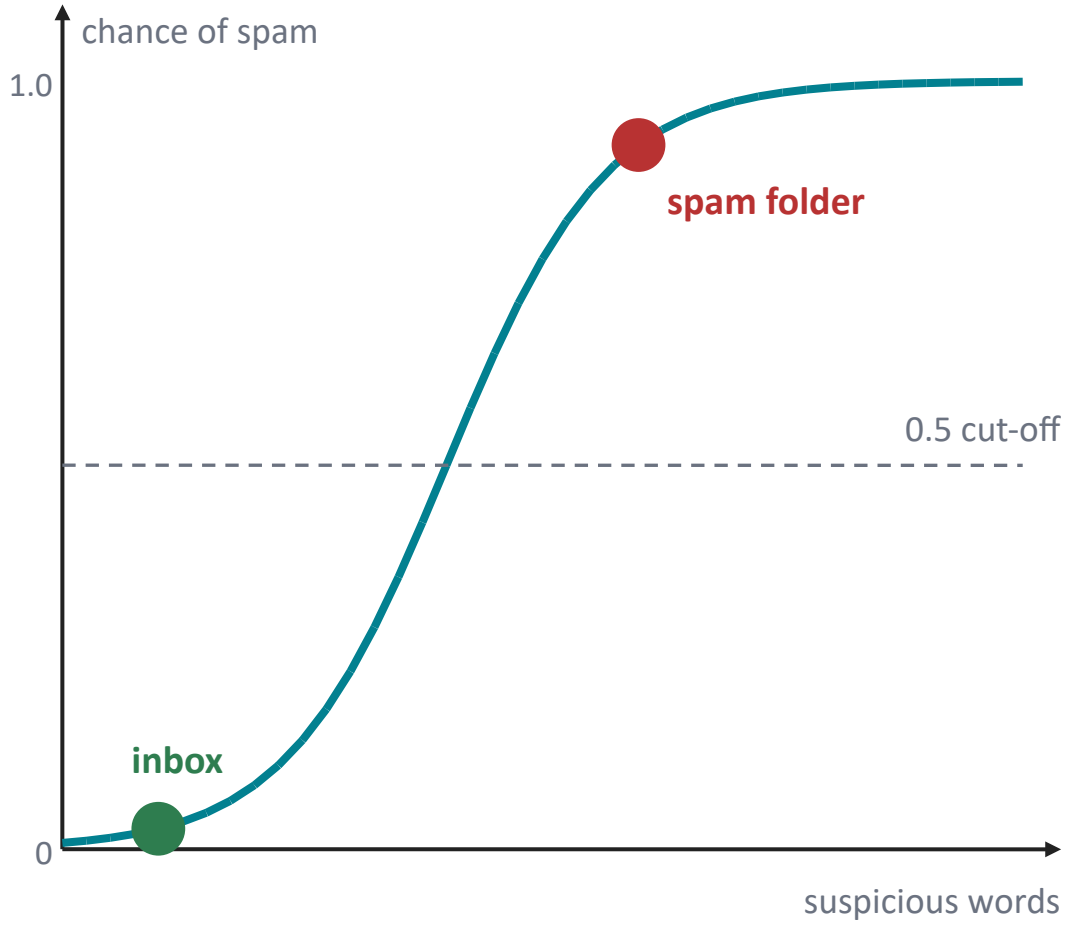
Model 2: Logistic Regression

Supervised · classification (yes / no)

What it does: despite its name, it answers yes/no questions. It gives a probability from 0 to 1, then uses a cut-off, usually 0.5.

How: it turns a score into an S-shaped curve of probability.

Simple example: number of suspicious words in an e-mail → chance of spam. 1 word: low chance → inbox. 6 words: high chance → spam folder.

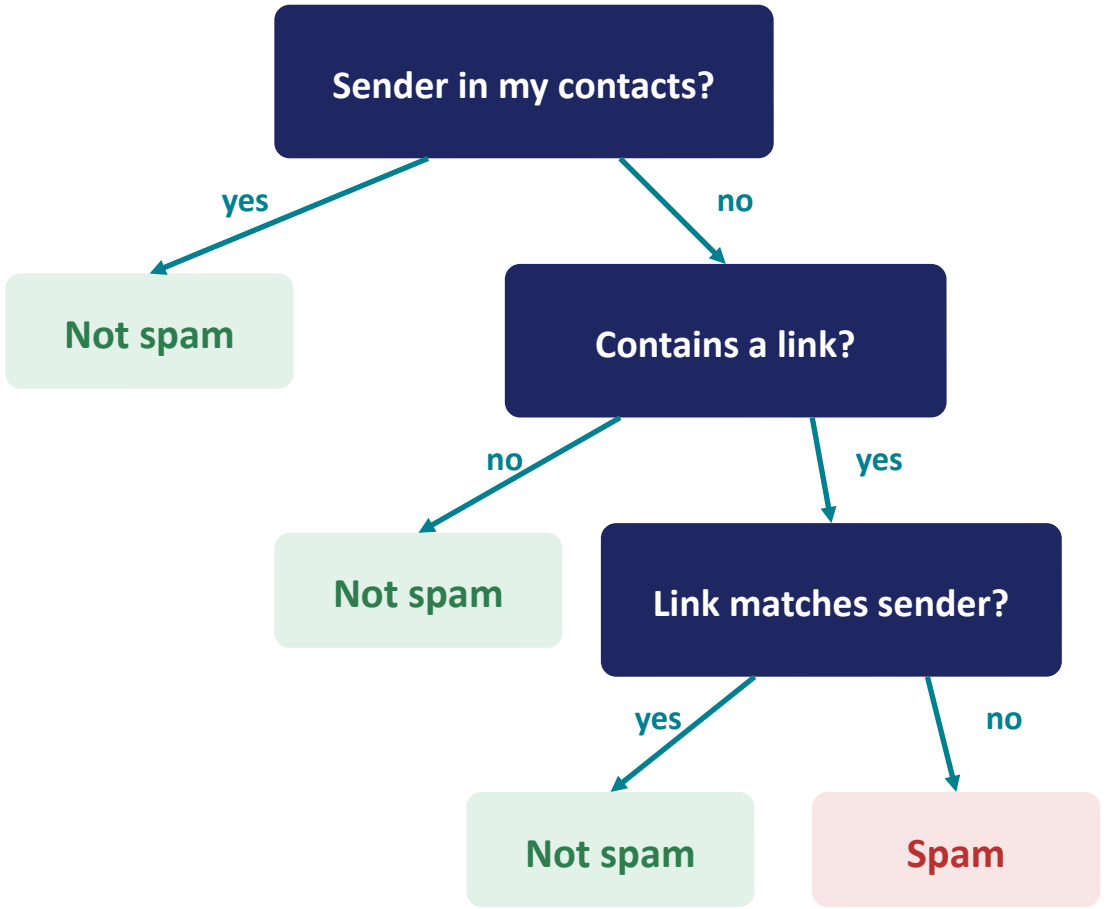


Model 3: Decision Tree

Supervised · classification or regression

What it does: asks a chain of simple yes/no questions, like a flowchart, until it reaches a decision.

How: from the training data it learns which question to ask first: the one that separates the examples best.



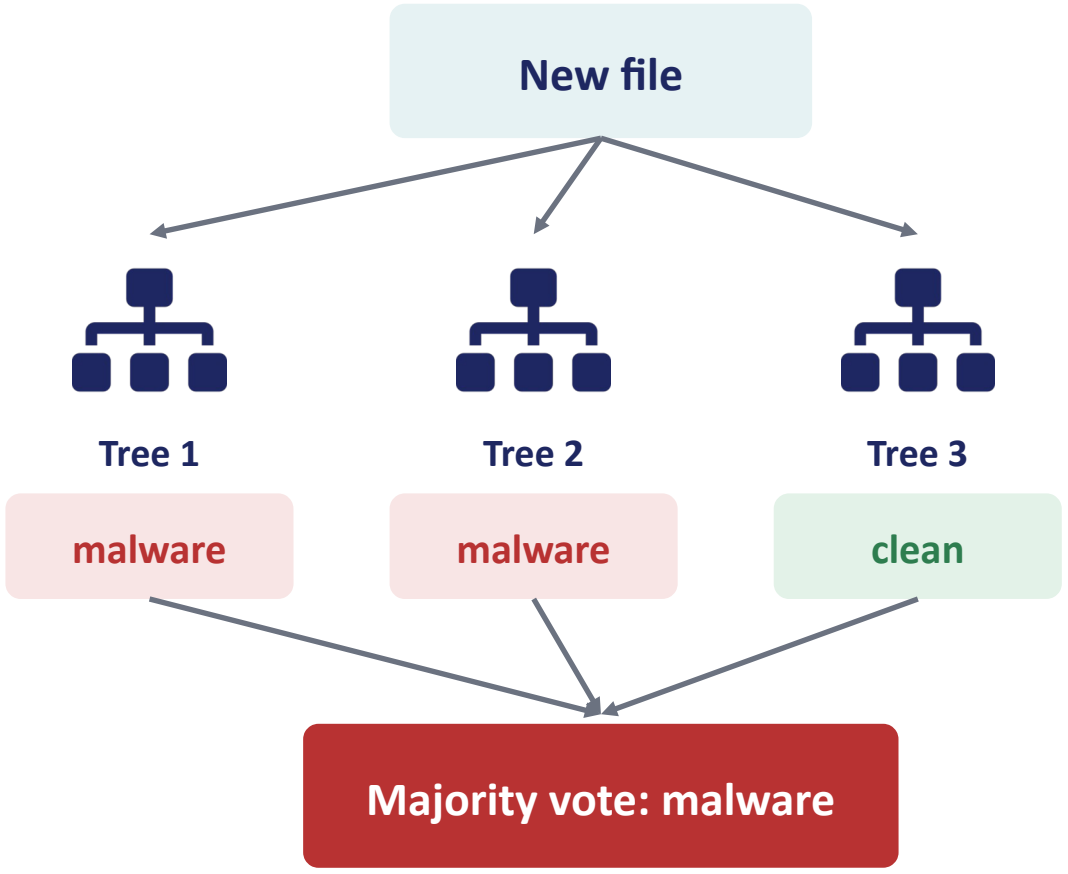
Simple example: is this e-mail spam? The tree on the right asks about the sender and the link. Easy to read and explain.

Model 4: Random Forest

Supervised · classification or regression

What it does: builds many different decision trees and lets them vote.

How: each tree learns from a slightly different sample of the data. The majority vote is the final answer, so it makes fewer mistakes than one tree.



Simple example: is this file malware? Tree 1: malware. Tree 2: malware. Tree 3: clean. Final answer: malware (2 of 3 votes).

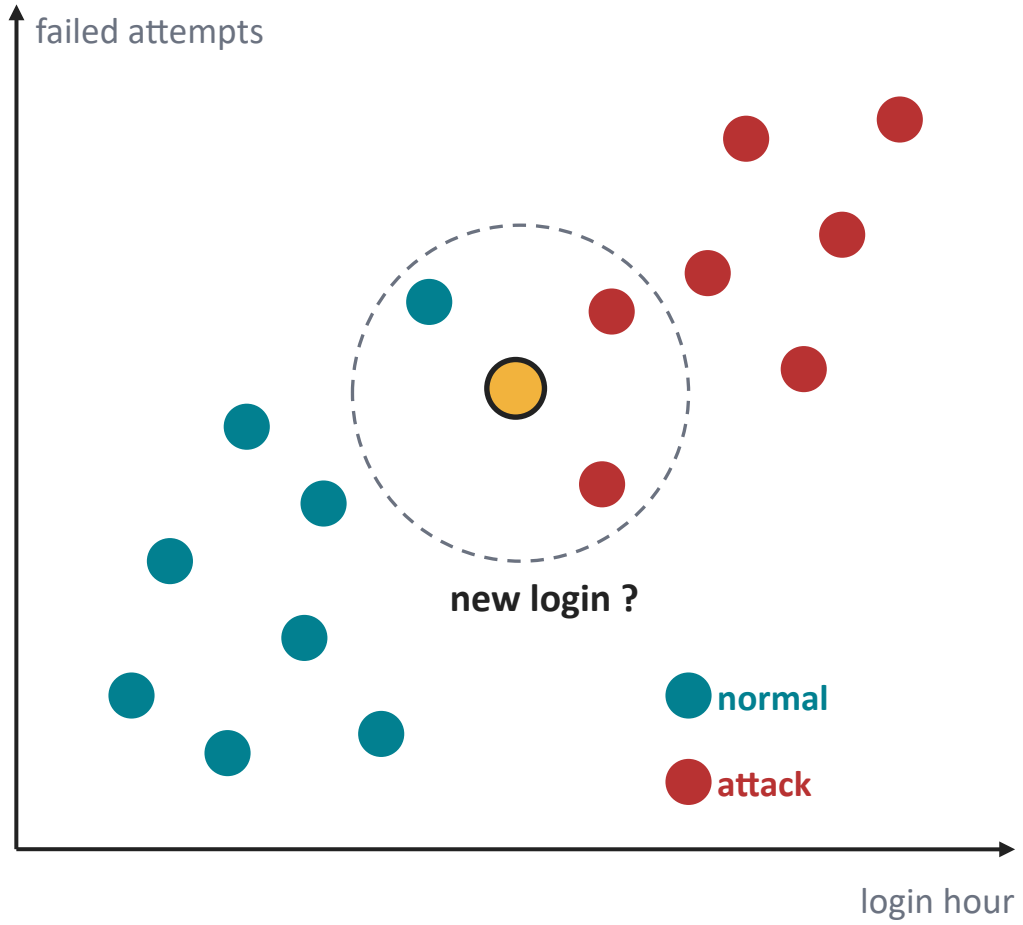
Model 5: k-Nearest Neighbours (k-NN)

Supervised · classification

What it does: labels a new item by looking at the k most similar items it has already seen, and takes a majority vote.

How: “similar” means close together on the chart. We choose k, for example k = 3.

Simple example: a new login. Its 3 nearest past logins: 2 were attacks, 1 was normal → the new login is labelled an attack.



Model 6: Naive Bayes

Supervised · classification

What it does: uses probabilities of clues. It learns how often each word appears in spam and in normal e-mail, then combines the clues to pick the most likely class.

How: “naive” because it treats every clue as independent of the others. Simple and fast.

Simple example: an e-mail with “winner” and “free” → spam is far more likely than normal.

How often each word appears

Word	in spam	in normal e-mail
winner	40%	1%
free	50%	5%
meeting	2%	20%

Illustrative values

New e-mail: “You are a winner! Claim your free gift”

↓

Clues point to: spam

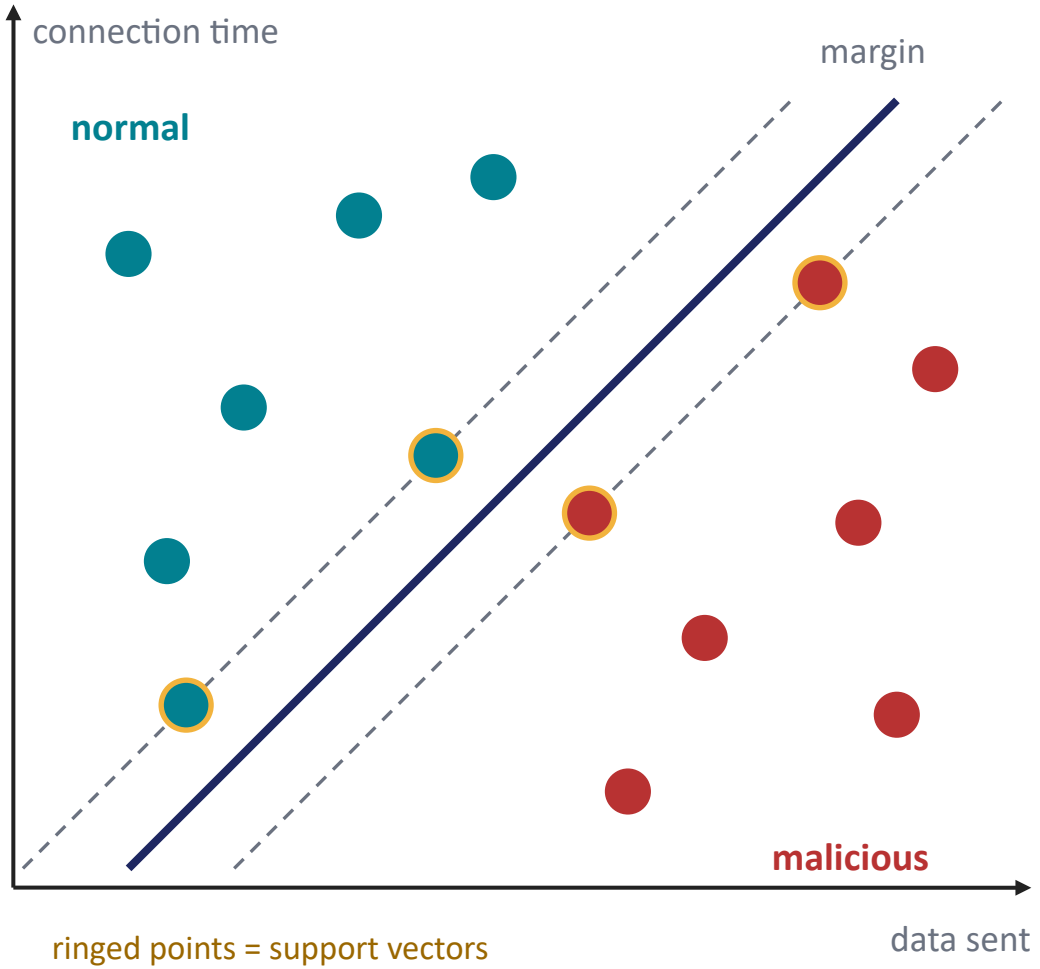
Model 7: Support Vector Machine (SVM)

Supervised · classification

What it does: draws the boundary that separates two classes with the widest possible gap (the margin).

How: only the points closest to the boundary, the support vectors, decide where it goes.

Simple example: separate normal and malicious network connections using two measurements, such as data sent and connection time.



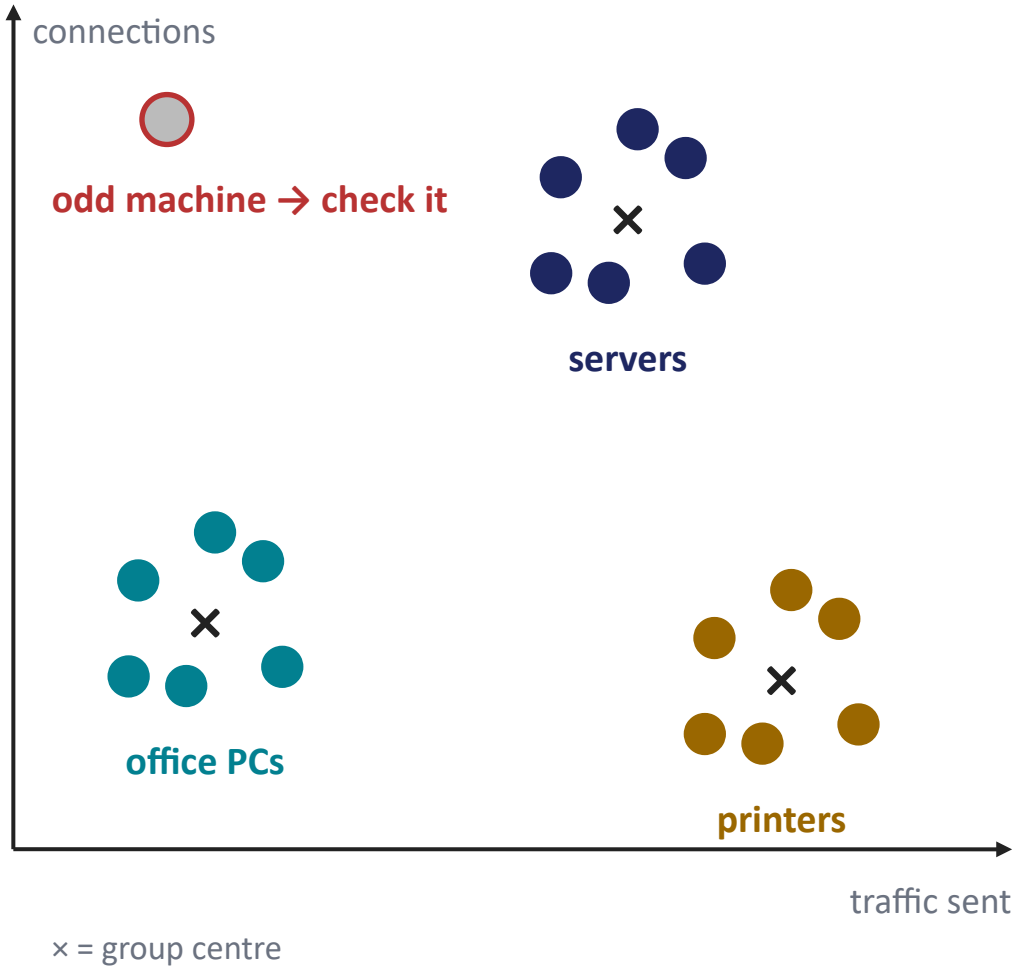
Model 8: k-Means Clustering

Unsupervised · clustering (no labels)

What it does: splits the data into k groups of similar items without any right answers.

How: 1) place k centres; 2) put each point with its nearest centre; 3) move each centre to the middle of its group; repeat.

Simple example: group the computers on a network by behaviour: office PCs, servers, printers. One machine fits no group → investigate it.



Choosing a Model: Quick Guide

Model	Learning type	Answer	Simple example
Linear regression	Supervised	Number	Exam mark from hours studied
Logistic regression	Supervised	Yes / no (probability)	Chance that an e-mail is spam
Decision tree	Supervised	Class or number	Yes/no questions about sender and link
Random forest	Supervised	Class or number	Many trees vote: malware or clean
k-nearest neighbours	Supervised	Class	New login labelled like its 3 nearest
Naive Bayes	Supervised	Class	Word clues such as “winner” and “free”
Support vector machine	Supervised	Class	Widest gap between normal and malicious
k-means	Unsupervised	Groups	Group network computers; find the odd one
Neural network	Supervised (mostly)	Class or number	Face unlock (next: deep learning)

Summary of Lecture 1

AI is a field of computer science; most AI today follows the rational-agent view.

Symbolic AI used rules written by people; machine learning learns the rules from data.

A program learns when its performance on a task improves with experience (T, E, P).

Three ways to learn: supervised, unsupervised, reinforcement.

Models: linear and logistic regression, decision tree, random forest, k-NN, naive Bayes, SVM, k-means.

Challenge Question (Optional)

For students who want to go further

A spam filter is 99% accurate, but almost all of its mistakes are phishing e-mails from look-alike bank domains.

Is it a good filter?

Which model from this lecture would let an analyst see why each e-mail was flagged?